

ONKAR KAJROLKAR

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SUMMARY

Results-driven Robotics and Automation Engineer with hands-on experience in robotic deployment, industrial automation, and system integration across manufacturing and production environments. Skilled in configuring, maintaining, and optimizing multi-axis robots, 3D-vision systems, and PLC-controlled systems. Background includes deploying vision-guided robotic arms, IoT-connected platforms, and AGVs, with a strong foundation in field engineering and cross-functional collaboration.

EDUCATION

M.S., Robotics and Autonomous Systems

Arizona State University, Tempe, Arizona

May 2025

GPA – 4.0/4.0

Relevant Coursework: Industrial Robots, Robotic Systems, Mechatronics Device Innovation.

B.E., Electronics Engineering

Mumbai University, Mumbai, India

Oct 2020

CGPA – 8.02/10

Relevant Coursework: Robotics, Control Systems, Embedded Systems, IoT.

TECHNICAL SKILLS

Programming: ROS/ROS2, SLAM navigation, OpenCV, C, C++, Python, Java, SQL, PLCs, IK Solvers

Platforms: RoboGuide, Universal Robots, MATLAB, Simulink, SolidWorks, AutoCAD

Industrial Systems: FANUC, ABB, SCARA, Allen-Bradley PLCs, Ladder Logic, Studio 5000, Siemens TIA Portal

Certifications: Industrial IoT, PLC Programming

PROFESSIONAL EXPERIENCE

Automation Intern, Seaboard Foods LLC, Oklahoma

Jul 2025 – Present

- Developed and deployed a neural-network computer vision system for real-time unit counting from a live production-line video feed, achieving 97% detection accuracy for continuous monitoring.
- Designed a plant-wide automation asset registry covering 150+ PLCs, HMIs and I/O modules, mapping network architecture and hardware locations to improve infrastructure visibility across engineering teams.
- Optimized robotic cutting cells with 3D vision and servo actuators, refining calibration across 3 production stations, improving alignment precision by 18%.
- Implemented PLC ladder logic updates and HMI control improvements across 4 automated processing units, stabilizing machine sequencing and reducing control-cycle latency by 20%.
- Improved vision-system performance by refining preprocessing and detection thresholds, increasing detection consistency by 95% across 50k+ processed units.
- Diagnosed and resolved solutions across mechatronic subsystems (pneumatics, drives, safety interlocks) during system interventions, restoring production operations 30% faster.

Robotics Engineer Intern, EntreVita Inc., Michigan

Jul 2024 – May 2025

- Integrated the MyCobot 280 M5 robotic arm into a ROS2 stack, building simulation and visualization workflows in Gazebo and RViz to support development and testing of manipulation tasks.
- Improved motion planning efficiency by 37% using MoveIt2 Task Constructor, benchmarking planners including OMPL, CHOMP, and STOMP while debugging open-source ROS pick-and-place pipelines.
- Configured a vision-guided pick-and-place system on a Jetson Nano, achieving 90% object recognition accuracy across 50 categories using CV techniques including edge detection, contour analysis, and template matching.
- Set up a remote robotics development environment using an Azure VM and SSH-connected Jetson Nano, enabling remote testing, monitoring, and control of the robotic system.
- Designed testing workflows in Gazebo simulation to validate perception and motion planning pipelines before hardware deployment, reducing physical robot debugging cycles by ~30%.

Survey Programmer, IPSOS Data Service Pvt. Ltd., Thane, India

Jun 2022 – Jun 2023

- Led a cross-functional team of 7 to implement advanced web development techniques and sentiment analysis using machine learning algorithms including KNN and Logistic Regression.
- Built data collection and processing pipelines in IBM Dimensions, structuring 10k+ survey responses into analyzable datasets, providing actionable insights for client strategy.
- Designed dynamic survey logic and automated validation rules, improving data integrity and reducing manual data cleaning by 25%.

- Implemented SAP S/4HANA Sales and Distribution (SD) workflows supporting enterprise supply chain operations for a global semiconductor client.
- Automated transaction workflows handling 1M+ daily financial and sales records, improving operational throughput and system reliability.
- Created Functional Designs, test cases, and documentation using ABAP programming while managing complex user queries and system enhancements.
- Developed ABAP functional designs and system enhancements, resolving 100+ production issues across the SAP SD.

PROJECTS**Writing with MyCobot Robotic Arm and its Digital Twin**

Aug 2023 – Nov 2023

- Orchestrated the integration of the MyCobot robotic arm with MATLAB, establishing a robust communication link for synchronized letter printing between the physical robot and its digital simulation.
- Developed a digital twin for the MyCobot robotic arm using SolidWorks and the arm's URDF file, enabling precise simulation and control via Simulink.

Autonomous line-following UAV (Unmanned Aerial Vehicle)

Jan 2024 – May 2024

- Designed and developed an autonomous line-following UAV using MATLAB and Simulink, integrating control systems to achieve precise line tracking and landing on a moving platform.
- Implemented advanced image processing algorithms for line detection and sensor fusion techniques with inputs from a camera, gyroscope, and accelerometer to enhance situational awareness and flight precision.

ROS-based Autonomous UGV (Unmanned Ground Vehicle)

Jan 2024 – May 2024

- Collaborated to develop an autonomous human-following robot, integrating an advanced occluded person detection algorithm and LiDAR-based obstacle avoidance to ensure safe, reliable tracking in dynamic environments.
- Led the implementation of the occluded person detection system, utilizing RGBD camera input to identify and follow a person with high accuracy, even in partially occluded scenarios.
- Designed and tested a LiDAR obstacle avoidance mechanism, achieving real-time performance in varied environments without compromising the safety or efficiency, demonstrating the robot's capability to update its trajectory.

Soft Robotic Arm for Elbow Muscle Hypertonicity Rehabilitation

Jan 2024 – May 2024

- Engineered a soft robotic arm specifically for aiding patients with elbow muscle hypertonicity, focusing on user-centric design and functionality.
- Utilized advanced soft robotics technology to create a flexible yet robust arm mechanism that aids in extending the patient's arm, thereby improving their range of motion.
- Incorporated electromyography (EMG) sensors to autonomously monitor and respond to the patient's muscle state, ensuring tailored assistance and adaptive motion control.

Industrial Robot Workcell Design and Simulation

Aug 2024 – Dec 2024

- Designed and simulated workcells in FANUC Roboguide for industrial robots, including FANUC ER-4iA, CRX-10iA cobot, and SCARA, with complete setup of user and tool frames.
- Configured PLC logic to simulate conveyor and safety interlocks as part of workcell validation.
- Executed projects involving advanced tasks such as cylindrical object creation using the CRX-10iA with a lathe machine (optimized cycle times from 250 to 1000), palletizing objects from conveyor belts, and complex pick-and-place operations with varying object sizes and orientations.
- Integrated safety protocols and simulated operator checkup routines to comply with industrial standards.

Dynamic Multi-Robot Patrolling and Monitoring in 3D Environments

Aug 2025 – Dec 2024

- Designed and implemented a two-level coordination strategy for multi-robot systems to minimize idleness and resolve spatial conflicts in 3D-patrolling tasks. Utilized global and local planners for target prioritization and obstacle avoidance.
- Simulated the system in Gazebo with a custom controller, integrating path planning, real-time conflict resolution, and dynamic adaptability to achieve uniform workspace coverage and stability.
- Conducted extensive evaluations, analyzing metrics like average idleness and conflict patterns, to optimize the system's efficiency and scalability for real-world surveillance applications.

PUBLICATIONS**Autonomous Car with Optimized Dynamic Range Using Machine Learning: A Review**

Jan 2020 – Oct 2020

- Spearheaded research in autonomous vehicle technologies, focusing on machine learning applications and optical sensor enhancement to address key challenges and integrate self-driving cars into smart city infrastructures. ([Link](#))